

Robust backstepping attitude tracking control of an underactuated spacecraft with saturation and time-variant perturbations

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Abstract

This study investigated associations of attitude tracking control of an underactuated spacecraft with consideration of saturation and perturbations. A nonsingular attitude tracking control was proposed which did not need limiting initial conditions of the quaternions. The controller was analyzed based on Lyapunov criteria and LaSalle's invariance theorem in the large-angle maneuver. In order to control, the complete kinematic and dynamic model of the underactuated spacecraft was reconstructed. According to simulation results, our controller has excellent robustness against the hard saturation, external disturbances, time-varying inertia uncertainties, and internal disturbances of actuators. As result, we found that the attitude controller was asymptotically stable under the soft saturation and the perturbations so that quaternions and angular velocity converged to the desired path within the 80 s. Also, it was still asymptotic stable under the hard saturation whose level is equal to 0.035 Nm, 3.5% of the soft saturation level. In this case, errors of quaternions and angular velocity were converged to the origin within the 150 s. Finally, the closed-loop system was verified by Adams-MATLAB co-simulation. The maximum verification errors for quaternions were less than 19%, while the maximum verification errors for angular velocity were less than 13.5%.

Keywords

Underactuated spacecraft, attitude tracking control, backstepping control, time-varying perturbation, saturation

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Introduction

Generally, three-axis attitude control of a spacecraft requires at least three actuators for all independent directions. But sometimes, due to design or failure events, the number of actuators is reduced from three, and in this case, the spacecraft becomes an underactuated system. Due to fewer actuators than degrees of freedom, the attitude control of an underactuated spacecraft is a crucial issue because of difficulty to achieve controllability and stability conditions. Also in presence of disturbances and uncertainties, as a result from the underactuation, actuator saturation plays a major part in the control design. Model uncertainties associated with actuator saturations lead to more complexity for control design of underactuated spacecraft. In this regard, we have not yet found an article on the attitude tracking control of the underactuated spacecraft in presence of uncertainties and saturations.

Based on literature survey, all research studies on the attitude control of the underactuated spacecraft can be summarized into three categories: dynamic, kinematic, and coupled kinematic–dynamic model. In the dynamic model, the control objective is the angular velocity. Also, the control input is the torque generated by actuators. But,

the purpose of the kinematic model is to control and stabilize the attitude parameters, and the angular velocity is virtual input. Finally, in the last category, the coupled kinematic–dynamic model is considered the cascade system. In this view, both the angular velocity and the attitude parameters are controlled by the actuator torques. In this point of view, few reports exist on the attitude tracking control of the underactuated spacecraft, and more works are focused on the regulation problems.

When only two control inputs are available, the angular velocity is globally stable controlled using the feedback control by outbib¹ and andriano.² Reyhanoglu³ claimed that the angular velocity of a rigid body using two control torques could not be exponentially stable by continuous feedback, and it provides a discontinuous feedback for this

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purpose. The angular velocity of an underactuated spacecraft is asymptotically stable by two independent torques.^{4,5} Moreover, angular velocity of underactuated spacecraft with uncertainty is stabilized by using continuous time-variant feedback, and the controller robustness has also been investigated.⁶ In this regard, many other articles have investigated the angular velocity control of the underactuated spacecraft, which is referred by Tsiotras and Doumchenko.⁷ Morin⁸ investigated the robustness of the angular velocity control for an underactuated spacecraft with respect to errors in the location of the actuators. The three-axis stabilization is proven by considering zero of total angular momentum.⁹ Also, local stability is achieved for a specific range of initial conditions. The spin stabilization of the underactuated spacecraft based on the Lyapunov criteria is provided by two the reaction wheels.¹⁰ The angular velocity control of the symmetric satellite using two torques is designed without measuring of the angular velocity.¹¹ The global asymptotic stability is achieved by LaSalle's invariance principle. In some articles, the dynamic model of the underactuated spacecraft is unknown, and then, heuristic algorithms such as neural network are used. Zeng¹² used deterministic learning to develop the controller for the underactuated satellite whose dynamic model is the black box. Also, many researchers control the underactuated spacecraft using the optimum algorithms.^{13–16} Also, the underactuated spacecraft is stabilized by using model predictive control approach.¹⁷ A robust control by using the H_∞ method is introduced by Han et al.¹⁸ such that the accuracy of the pointing in the presence of the disturbances is improved.

In the point of view of the kinematic model, Endo et al.¹⁹ proposed the attitude control based on the input–output linearization method. The method uses the error compensator to increase robustness. Also, the generalized inversion method for attitude control of underactuated spacecraft is presented.²⁰ The singular controller is designed using the kinematic–dynamic model approach. Furthermore, Gui et al.^{21,22} introduced a singular attitude controller for the underactuated spacecraft based on the generalized inversion method. Cao and Chen²³ presented a new sliding mode controller for the attitude control and determination systems of an uncertain satellite. Their simulation results were performed to show the efficacy of the proposed controller for a full-actuated satellite.

According to literature review on the underactuated spacecraft, some research studies proposed a non-model-based controller that stability has not been analyzed. However, in the category of the model-based controller, few reports exist on the attitude tracking control with the coupled kinematic–dynamic model. Indeed, in this point of view, more works are concentrated on the regulation. Also, many controllers using the method of the model-based attitude tracking control are provided based on the singularity, and they require special initial conditions.^{4,14,19–22} An unresolved issue in the field of underactuated spacecraft control is the robust control of the cascade model with saturation and perturbations under the large attitude maneuver. The current research has some important

advantages over previous works. In this article, we consider the coupled kinematic–dynamic model (cascade system). In this view, both the angular velocity and attitude parameters are simultaneously controlled by two actuator torques. Furthermore, we design the attitude tracking control of an underactuated spacecraft in the large-angle maneuver based on Lyapunov criteria which has robustness against saturation and time-varying perturbations.

Our cascade model has two parts: actuated and unactuated part. At first, the unactuated part was stabilized by using a novel hypersurface. Afterward, the asymptotic stability was achieved by LaSalle's invariance theorem. Then, backstepping control was designed based on the Lyapunov direct method such that the asymptotic stability of the closed-loop system was guaranteed. Then, improving the controller robustness was obtained by a finite-time nonlinear disturbance observer (FNDO). Also, we modified FNDO-based backstepping (FNDO-BS) control to improve the robustness in the presence of the saturation. According to simulation results, the modified FNDO-BS (MFNDO-BS) controller has excellent robustness against hard saturation, external disturbances, time-varying inertia uncertainties, and internal disturbances of the actuators. Also, the closed-loop system was verified by Adams-MATLAB co-simulation. Our major contributions are presented in the following:

1. Design of a robust backstepping attitude tracking control for a large maneuver of an underactuated spacecraft which does not need limiting initial conditions of the quaternions.
2. Design of a MFNDO-BS controller which has excellent robustness against hard saturation, external disturbances, inertia uncertainties, and internal disturbances of the actuators, all of which are time-variant.
3. Stability analysis of the whole system including nonlinear observer and control based on Lyapunov criteria.
4. System was verified by Adams-MATLAB co-simulation.

This article is organized as follows. First, the cascade model of underactuated spacecraft is described. In the next section, an attitude tracking control is designed for underactuated spacecraft. Finally, system verification and the simulation results are presented.

Cascade model of underactuated spacecraft

As we know, whenever the exchange angular momentum devices such as reaction wheels are used in the body coordinate, equation (1) can be written²⁴ as

$$\{J_0 + \Delta J\}\dot{\omega} = -\dot{h}_w - \omega^\times \{J_0 + \Delta J\}\omega - \omega^\times h_w + T_d \quad (1)$$

where $h_w, \dot{h}_w, \omega, \dot{\omega}, T_d \in \mathbf{R}^3$ are the angular momentum of reaction wheels and its time derivative, the angular velocity and its time derivative, and the external disturbance

torque in the body coordinate, respectively. Also, $\mathbf{J}_0 \in R^{3 \times 3}$ is the symmetric inertia matrix which is the known matrix and $\Delta \mathbf{J} \in R^{3 \times 3}$ is the unknown time-varying matrix; and the operator $\omega^\times$ denotes skew-symmetric matrix acting on the vector $\omega = [\omega_x \ \omega_y \ \omega_z]^T$ and can be defined as follows

$$\omega^\times = \begin{bmatrix} 0 & -\omega_z & \omega_y \\ \omega_z & 0 & -\omega_x \\ -\omega_y & \omega_x & 0 \end{bmatrix} \quad (2)$$

Quaternions represent the spacecraft attitude in the inertial coordinate. Quaternion errors are defined by Nadafi et al.²⁵ as

$$\bar{\mathbf{q}}_e = \begin{bmatrix} q_{e0} \\ \mathbf{q}_e \end{bmatrix} = \begin{bmatrix} q_{d0} & q_{d1} & q_{d2} & q_{d3} \\ q_{d1} & -q_{d0} & -q_{d3} & q_{d2} \\ q_{d2} & q_{d3} & -q_{d0} & -q_{d1} \\ q_{d3} & -q_{d2} & q_{d1} & -q_{d0} \end{bmatrix} \begin{bmatrix} q_0 \\ q_1 \\ q_2 \\ q_3 \end{bmatrix} \quad (3)$$

where $\bar{\mathbf{q}}_d = [q_{d0} \ q_{d1} \ q_{d2} \ q_{d3}]^T$, $\bar{\mathbf{q}} = [q_0 \ q_1 \ q_2 \ q_3]^T$, and $\bar{\mathbf{q}}_e = [q_{e0} \ q_{e1} \ q_{e2} \ q_{e3}]^T$ are desired, spacecraft, and error attitude parameters, respectively.

Remark 1. Quaternion errors are constrained by the condition $q_{e0}^2 + \mathbf{q}_e^T \mathbf{q}_e = 1$.

According to equation (3), whenever $\bar{\mathbf{q}} = \bar{\mathbf{q}}_d$, then $\bar{\mathbf{q}}_e$ is equal to $[1 \ 0 \ 0 \ 0]^T$. The time derivative of $\bar{\mathbf{q}}_e$ is calculated using equation (4)

$$\dot{\bar{\mathbf{q}}}_e = \begin{bmatrix} \dot{q}_{e0} \\ \dot{\mathbf{q}}_e \end{bmatrix} = \begin{bmatrix} -\frac{1}{2} \omega_e^T \mathbf{q}_e \\ -\frac{1}{2} \omega_e^\times \mathbf{q}_e + \frac{1}{2} q_{e0} \omega_e \end{bmatrix} \quad (4)$$

As described by Cai et al.,²⁶ the error of the angular velocity is $\omega_e = \omega - \mathbf{C}\omega_d$ and its time derivative is obtained in the following form

$$\dot{\omega}_e = \dot{\omega} + \omega_e^\times \mathbf{C}\omega_d - \mathbf{C}\dot{\omega}_d \quad (5)$$

where ω_d is the desired angular velocity of spacecraft, and the rotation matrix from the desired coordinate to the body coordinate is given by $\mathbf{C} = \{q_{e0}^2 - \mathbf{q}_e^T \mathbf{q}_e\} \mathbf{I}_3 + 2\mathbf{q}_e \mathbf{q}_e^T - 2q_{e0} \mathbf{q}_e^\times$ where $\mathbf{I}_3$ is the 3×3 identity matrix.²⁷

Remark 2. Notice that $\|\mathbf{C}\| = 1$ and $\dot{\mathbf{C}} = -\omega_e \mathbf{C}$.

Assumption 1. We assume that the body coordinate is aligned to the direction of the principal axes, and the z -direction is the unactuated axis; then, $h_{w_z} = \dot{h}_{w_z} = 0$.

Taking into account the controllability condition of the rigid spacecraft equipped with two reaction wheels, total angular momentum should be zero^{27–29}; then, $J_z \omega_z$ must be zero. Therefore, the vector of the angular velocity error will be reduced to $\omega_e^r = [\omega_{ex} \ \omega_{ey}]^T$. As we know that the total angular momentum remains equal to the initial value at all times if reaction wheel is used. Thus, using detumbling phase, total angular momentum converges to zero. The practical tools of implementing this condition

are the magnetic actuators in the spacecraft, which are always installed in the spacecraft to desaturate the reaction wheel.

Thus, according to equations (1), (4), and (5), the cascade model is obtained as follows

$$\dot{\omega}_e^r = -\{\mathbf{J}_0^r + \Delta \mathbf{J}^r\}^{-1} \dot{\mathbf{h}}_w^r - \mathbf{C}^r \dot{\omega}_d^r + \{c_{31} \omega_{dx} + c_{32} \omega_{dy}\} [\omega_{ey} \ -\omega_{ex}]^T + \{\mathbf{J}_0^r + \Delta \mathbf{J}^r\}^{-1} \mathbf{T}_d^r \quad (6a)$$

$$\dot{\mathbf{q}}_e = \frac{1}{2} \begin{bmatrix} q_{e0} & -q_{e3} \\ q_{e3} & q_{e0} \\ -q_{e2} & q_{e1} \end{bmatrix} \omega_e^r \quad (6b)$$

where c_{ij} is the element of the rotation matrix $\mathbf{C}$. In equation (6a), each vector in the form of $\mathbf{O}^r \in R^2$ is equal to $[O_x \ O_y]^T$ based on components of vector $\mathbf{O} \in R^3$, and each matrix in the form of $\mathbf{P}^r \in R^{2 \times 2}$ is equal to

$$\begin{bmatrix} P_{xx} & P_{xy} \\ P_{yx} & P_{yy} \end{bmatrix} \text{ based on elements of matrix } \mathbf{P} \in R^{3 \times 3}.$$

Consider equations (6a) and (6b). The cascade model has two types of variables: actuated ω_e^r and unactuated variable $\mathbf{q}_e$.

Remark 3. We consider input vector $\mathbf{u}$ is equal to $-\{\mathbf{J}_0^r\}^{-1} \dot{\mathbf{h}}_w^r$.

Remark 4. $\mathbf{J}_0$ is a known bounded matrix and $\Delta \mathbf{J}$ is an unknown time-varying and bounded matrix. Also, time derivative of $\Delta \mathbf{J}$ is presumed bounded. We can write $\{\mathbf{J}_0^r + \Delta \mathbf{J}^r\}^{-1} = \{\mathbf{J}_0^r\}^{-1} - \{\mathbf{J}_0^r\}^{-1} \Delta \mathbf{J}^r (\mathbf{I}_2 + \{\mathbf{J}_0^r\}^{-1} \Delta \mathbf{J}^r)^{-1} \{\mathbf{J}_0^r\}^{-1}$ and $\Delta \mathbf{J}^r = \{\mathbf{J}_0^r\}^{-1} \Delta \mathbf{J}^r (\mathbf{I}_2 + \{\mathbf{J}_0^r\}^{-1} \Delta \mathbf{J}^r)^{-1} \{\mathbf{J}_0^r\}^{-1}$ where $\mathbf{I}_2 \in R^{2 \times 2}$ is the identity matrix.³⁰

Assumption 2. Assume ω_d^r and $\dot{\omega}_d^r$ are bounded continuous vector such that $\|\omega_d^r\| \leq \beta_1$ and $\|\dot{\omega}_d^r\| \leq \beta_2$.

Assumption 3. The external disturbance $\mathbf{T}_d^r$ concluding the torques of the solar radiation, the gravity gradient, the aerodynamics, magnetics, and other perturbations is unknown bounded and sufficient smooth vector. Also, $\dot{\mathbf{T}}_d^r$ is assumed bounded. Therefore, there exist constants $\eta_0, \eta_1 > 0$ such that

$$\|\mathbf{T}_d^r\| \leq \eta_0, \quad \|\dot{\mathbf{T}}_d^r\| \leq \eta_1 \quad (7)$$

Internal disturbance of the actuator

The internal disturbance of reaction wheels can have an undesirable effect on control system performances.³¹ Zhou et al.³² presented a control scheme for an uncertain satellite equipped with reaction wheels so that results showed an engineering value for improving reliability and prolonging on-orbit working lifetime of satellites. But this controller designed to stabilize a full-actuated spacecraft. Recently, the disturbance model of

reaction wheels is developed by some research.^{33–37} Internal disturbances of the reaction wheel include the flywheel imbalance, the bearing irregularity, the torque ripple, and cogging in the brushless DC motor.³⁵ According to the literature, in general, the flywheel imbalance has the largest portion of internal disturbances. Masrerson et al.³⁵ presented a full disturbance model for the reaction wheel in the form $\dot{h}_{dw} = K_w \omega_w^2 \sin(\varrho_w \omega_w)$, where $\dot{h}_{dw}$ and ω_w are the disturbance torque and the angular velocity of the reaction wheel, respectively; and K_w and ϱ_w are reaction wheel coefficients which must be determined theoretically or practically.

Assumption 4. Due to improvement of the robustness and the stability of the controller under the reaction wheel disturbance consideration, also in order to avoid the complexity of coefficient determinations, we assumed that the actuator disturbance $\dot{h}_{dw}$ had the form as follows

$$\dot{h}_{dw} = \rho_w \dot{h}_w \quad (8)$$

where $\rho_w \in [0 \ 1)$ is the unknown constant and $\dot{h}_w$ is the generated torque by the actuator. In other words, we suppose that the internal disturbance of the reaction wheel reduces the performance of the actuator. Consequently, according to equation (6a), Remarks 3 and 4, and Assumption 4, equation (9) is obtained as follows

$$\begin{aligned} \dot{\omega}_e^r = & \mathbf{u} + \{\mathbf{J}_0^r\}^{-1} \rho_w \dot{h}_w^r + \overline{\Delta \mathbf{J}^r} \{\mathbf{I}_2 - \rho_w\} \dot{h}_w^r - \mathbf{C}^r \dot{\omega}_d^r \\ & + \{c_{31} \omega_{dx} + c_{32} \omega_{dy}\} [\omega_{ey} \ -\omega_{ex}]^T \\ & + \{\mathbf{J}_0^r + \Delta \mathbf{J}^r\}^{-1} \mathbf{T}_d^r \end{aligned} \quad (9)$$

where $\mathbf{I}_2 \in R^{2 \times 2}$ is the identity matrix and $\rho_w \in R^{2 \times 2}$ is equal to $\text{diag}(\rho_{wx}, \rho_{wy})$.

Cascade model

As explained above, we can obtain the cascaded model for the attitude tracking maneuver of the underactuated spacecraft with external disturbances, actuator disturbances, and uncertainties by using simple algebraic transformation as follows

$$\dot{\omega}_e^r = \mathbf{u} + \mathbf{F} + \mathbf{d} \quad (10a)$$

$$\dot{\mathbf{q}}_e = \frac{1}{2} \begin{bmatrix} q_{e0} & -q_{e3} \\ q_{e3} & q_{e0} \\ -q_{e2} & q_{e1} \end{bmatrix} \omega_e^r = \mathbf{A}(\mathbf{q}_e) \omega_e^r \quad (10b)$$

where $\mathbf{d}$ is the total perturbation including external disturbances, time-varying inertia uncertainties, and internal disturbances of actuators. Also, according to diagonal matrix ρ_w and $\mathbf{A}(\mathbf{q}_e) \in R^{3 \times 2}$, we write

$$\mathbf{F} = -\mathbf{C}^r \dot{\omega}_d^r + \{c_{31} \omega_{dx} + c_{32} \omega_{dy}\} [\omega_{ey} \ -\omega_{ex}]^T \quad (11a)$$

$$\mathbf{d} = -\rho_w \mathbf{u} - \overline{\Delta \mathbf{J}^r} \{\mathbf{I}_2 - \rho_w\} \mathbf{J}_0^r \mathbf{u} + \{\mathbf{J}_0^r + \Delta \mathbf{J}^r\}^{-1} \mathbf{T}_d^r \quad (11b)$$

where $\mathbf{F} \in R^2$ is a known differentiable vector.

Corollary 1. $\mathbf{d}, \dot{\mathbf{d}} \in$ are unknown bounded vectors.

Proof. According to controllability definition, we know that $\mathbf{u}$ and then $\dot{\mathbf{u}}$ must be bounded. Also, given Assumptions 3–4 and Remark 4, $\rho_w, \mathbf{J}_0^r, \Delta \mathbf{J}^r, \mathbf{T}_d^r$, and their time derivative are bounded. As a result, due to equation (11b), $\mathbf{d}$ and $\dot{\mathbf{d}}$ are bounded. $\square$

Control objective

In this article, the perfect attitude tracking is achieved if both actuated variable ω_e^r and unactuated variable $\mathbf{q}_e$ of the underactuated spacecraft equipped with two reaction wheels converge to origin in the presence of perturbations and saturation.

Lemma 1. Consider the cascade model presented in equation (10a); according to above Remarks and Assumptions, the following FNDO is proposed to estimate $\mathbf{d}$ as follows³⁸

$$\begin{aligned} \dot{\chi}_0 &= \mathbf{v}_0 + \mathbf{u} + \mathbf{F} \\ \mathbf{v}_0 &= -\kappa_0 \text{diag}(\mathbf{L}^{1/2} |\chi_0 - \omega_e^r|^{1/2}) \text{sign}(\chi_0 - \omega_e^r) + \chi_1 \\ \dot{\chi}_1 &= -\kappa_1 \mathbf{L} \text{sign}(\chi_1 - \mathbf{v}_0) \end{aligned} \quad (12)$$

$\kappa_i > 0, i = 0, 1$ and $\mathbf{L} \in R^2$ must be designed; $\chi_0 \in R^2$ and $\chi_1 \in R^2$ are the estimation of ω_e^r and $\mathbf{d}$, respectively. Also, $\mathbf{v}_0 \in R^2$ is an auxiliary vector.

Remark 8. For every $\mathbf{Y} \in R^n$, $\text{sign}(\mathbf{Y})$ is defined as $[\text{sign}(y_1) \ \dots \ \text{sign}(y_n)]^T$. Consider Lipschitz coefficients $L_j \geq \dot{d}_j$ with $j = x, y$, $\text{diag}(\mathbf{L}^{1/2} |\chi_0 - \omega_e^r|^{1/2}) = \text{diag}(L_x^{1/2} |\chi_{0x} - \omega_{ex}|^{1/2}, L_y^{1/2} |\chi_{0y} - \omega_{ey}|^{1/2})$.

The FNDO error is governed in the form of

$$\begin{aligned} \dot{e}_{\chi_0} &= \dot{\chi}_0 - \dot{\omega}_e^r = -\kappa_0 \text{diag}(\mathbf{L}^{1/2} |e_{\chi_0}|^{1/2}) \text{sign}(e_{\chi_0}) + e_{\chi_1} \\ \dot{e}_{\chi_1} &= \dot{\chi}_1 - \dot{\mathbf{d}} = -\kappa_1 \mathbf{L} \text{sign}(e_{\chi_1} - \dot{e}_{\chi_0}) - \dot{\mathbf{d}} \end{aligned} \quad (13)$$

where $e_{\chi_0} = \chi_0 - \omega_e^r$ and $e_{\chi_1} = \chi_1 - \mathbf{d}$. FNDO errors e_{χ_0} and e_{χ_1} can converge to zero in finite time by selecting suitable parameters κ_i and L_j .³⁸ In fact, if we consider the Lyapunov function $V_{e_x} = \beta_{\chi_0} \|e_{\chi_0}\|^2 + \beta_{\chi_1} \|e_{\chi_1}\|^2$ for FNDO, the inequality $\dot{V}_{e_x} \leq -V_{e_x}$ can be established. β_{χ_0} and β_{χ_1} are positive constants. Accordingly, Zeng,¹² Qinglei et al.,³⁰ and Ma et al.³⁹ applied FNDO for the attitude control of the full-actuated satellite in the presence of perturbation, but in this article, the FNDO-based controller is innovatively designed to control the underactuated spacecraft.

Controller design

In this section, we propose a novel hypersurface as a backstepping technique to achieve the attitude tracking control of the underactuated spacecraft. For this purpose, we stabilize unactuated variables via innovative virtual inputs, and then, all state variables are controlled asymptotically stable by using two torques.

Hypersurface design

Unactuated variables can be asymptotically stabilized by innovative hypersurfaces $\omega_e^r = \Phi(q_e)$, with $\Phi(0) = 0$, such that the origin of equation (10b) is asymptotically stable. Indeed, the virtual input is considered as follows

$$\begin{aligned}\omega_e^r &= \Phi(q_e) \\ \Phi(q_e) &= [\varphi_1 \quad \varphi_2]^T\end{aligned}\quad (14)$$

Theorem 1. Consider equations (10b) and (12); choosing $\Phi = -2[aq_{e2}q_{e3} + q_{e0}q_{e1}\lambda_1 \quad bq_{e1}q_{e3} + q_{e0}q_{e2}\lambda_2]^T$ guarantees unactuated variables converge to origin, where $\lambda_i > 0$, $a = \lambda_2 - \lambda_3$, and $b = \lambda_3 - \lambda_1$.

Proof. For this purpose, the Lyapunov function is supposed as follows

$$V_a = \frac{1}{2}q_e^T A q_e \quad (15)$$

where V_a is the positive definite and $A \in R^{3 \times 3}$ is as follows

$$A = \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{bmatrix} \quad \lambda_i > 0, \quad i = 1, 2, 3 \quad (16)$$

Taking the time derivative of V_a and applying equations (10b) and (14), $\dot{V}_a$ becomes

$$\dot{V}_a = q_e^T A \dot{q}_e = q_e^T A A(q_e) \omega_e^r = q_e^T A A(q_e) \Phi \quad (17)$$

The following equation can easily be obtained from equations (10b), (16), and (17)

$$\begin{aligned}\dot{V}_a &= 0.5\varphi_1 q_{e0}q_{e1}\lambda_1 - 0.5\varphi_2 q_{e3}q_{e1}\lambda_1 + 0.5\varphi_1 q_{e2}q_{e3}\lambda_2 \\ &\quad + 0.5\varphi_2 q_{e0}q_{e2}\lambda_2 - 0.5\varphi_1 q_{e1}q_{e3}\lambda_3 + 0.5\varphi_2 q_{e1}q_{e3}\lambda_3\end{aligned}\quad (18)$$

By sorting, we can write

$$\begin{aligned}\dot{V}_a &= 0.5\varphi_1 (q_{e0}q_{e1}\lambda_1 + a q_{e2}q_{e3}) + 0.5\varphi_2 (q_{e0}q_{e2}\lambda_2 \\ &\quad + b q_{e1}q_{e3}) \\ a &= \lambda_2 - \lambda_3 \\ b &= \lambda_3 - \lambda_1\end{aligned}\quad (19)$$

$\Phi(q_e)$ is a nonlinear function of q_e ; then, one of the choices of Φ can be as follows

$$\begin{aligned}\varphi_1 &= -2(aq_{e2}q_{e3} + q_{e0}q_{e1}\lambda_1) \\ \varphi_2 &= -2(bq_{e1}q_{e3} + q_{e0}q_{e2}\lambda_2)\end{aligned}\quad (20)$$

Remark 5. According to Remark 1, we know that q_{e0} is dependent on q_e ; thus, obviously, Φ is a function of q_e .

Using equation (20) and substituting this into equation (19) yields

$$\begin{aligned}\dot{V}_a &= -(aq_{e2}q_{e3})^2 - (q_{e0}q_{e1}\lambda_1)^2 - (bq_{e1}q_{e3})^2 \\ &\quad - (q_{e0}q_{e2}\lambda_2)^2 - 2a\lambda_1 q_{e0}q_{e1}q_{e2}q_{e3} - 2b\lambda_2 q_{e0}q_{e1}q_{e2}q_{e3}\end{aligned}\quad (21)$$

Remark 6. Presume $-a\lambda_1 = b\lambda_2$; then, due to equation (19), we have $\lambda_1 = \lambda_2$ and $a = -b$.

By Remark 6, equation (21) can be put in the following form

$$\dot{V}_a = -a^2 q_{e3}^2 \{q_{e1}^2 + q_{e2}^2\} - \lambda_1^2 q_{e0}^2 \{q_{e1}^2 + q_{e2}^2\} \quad (22)$$

In accordance with Remark 1, we can write

$$\begin{aligned}\dot{V}_a &= -a^2 q_{e3}^2 \{q_{e1}^2 + q_{e2}^2\} - \lambda_1^2 \{q_{e1}^2 + q_{e2}^2\} \{1 - q_{e3}^2 \\ &\quad - (q_{e1}^2 + q_{e2}^2)\} \\ &= -a^2 q_{e3}^2 \{q_{e1}^2 + q_{e2}^2\} \\ &\quad - \lambda_1^2 \{q_{e1}^2 + q_{e2}^2\} \{1 - (q_{e1}^2 + q_{e2}^2)\} \\ &\quad + \lambda_1^2 \{q_{e1}^2 + q_{e2}^2\} q_{e3}^2 \\ &= -\{a^2 - \lambda_1^2\} q_{e3}^2 \{q_{e1}^2 + q_{e2}^2\} \\ &\quad - \lambda_1^2 \{q_{e1}^2 + q_{e2}^2\} \{1 - (q_{e1}^2 + q_{e2}^2)\}\end{aligned}\quad (23)$$

Remark 7. Consider $|a| \leq |\lambda_1|$.

With respect to $\|q_e\| \leq 1$, then $q_{e1}^2 + q_{e2}^2 \leq 1$, and given Remark 7, it is clear that $\dot{V}_a$ is negative semidefinite. In order to guarantee the asymptotic stability of origin, $q_e = 0$, we can use LaSalle's invariance principle to show that equation (23) is zero only at the origin. Given equation (23), $\dot{V}_a$ can be zero in two possible cases:

Case I. $q_{e1}^2 + q_{e2}^2 = 0$:

For this case, let $q_{e1}(t) = q_{e2}(t) = 0$ be a solution of $\dot{V}_a = 0$; then, according to LaSalle's invariance principle, the time derivative of $q_{e1}(t)$ and $q_{e2}(t)$ is zero. Then, due to equation (10b) and $\omega_e^r = [\omega_{ex} \quad \omega_{ey}]^T$, we have

$$\begin{aligned}\dot{q}_{e1} &= 0 = 0.5q_{e0}\omega_{ex} - 0.5q_{e3}\omega_{ey} \\ \dot{q}_{e2} &= 0 = 0.5q_{e3}\omega_{ex} + 0.5q_{e0}\omega_{ey}\end{aligned}\quad (24)$$

Therefore

$$\begin{aligned}&-0.5q_{e0}\omega_{ey}\omega_{ex} + 0.5q_{e3}\omega_{ey}^2 + 0.5q_{e3}\omega_{ex}^2 + 0.5q_{e0}\omega_{ey}\omega_{ex} \\ &= 0.5q_{e3}(\omega_{ex}^2 + \omega_{ey}^2) = 0\end{aligned}\quad (25)$$

It is easily proved that q_{e3} is equal to zero; therefore, $\dot{V}_a$ can only be zero at the origin in this case.

Case II. $q_{e1}^2 + q_{e2}^2 = 1$ and $q_{e3} = 0$:

In this case, as described in Remark 1, $q_{e0}(t) = q_{e3}(t) = 0$ is a solution of $\dot{V}_a = 0$, and due to LaSalle's invariance principle, the time derivative of $q_{e0}(t)$ and $q_{e3}(t)$ is zero too; thus, we can write

$$\begin{aligned}\dot{q}_{e0} &= 0 = -0.5q_{e1}\omega_{ex} - 0.5q_{e2}\omega_{ey} \\ \dot{q}_{e3} &= 0 = -0.5q_{e2}\omega_{ex} + 0.5q_{e1}\omega_{ey}\end{aligned}\quad (26)$$

Therefore

$$\begin{aligned} & -0.5q_{e1}\omega_{ex}^2 - 0.5q_{e2}\omega_{ex}\omega_{ey} + 0.5q_{e2}\omega_{ex}\omega_{ey} - 0.5q_{e1}\omega_{ey}^2 \\ & = -0.5q_{e1}(\omega_{ex}^2 + \omega_{ey}^2) = 0 \end{aligned} \quad (27)$$

Thus, $q_{e1} = 0$ and also we have $q_{e2} = 0$. Thus, in this case, we can declare that $q_{e1}^2 + q_{e2}^2 = 1$ cannot be established whenever $q_{e3} = 0$; and therefore, it is proved that $\dot{V}_a$ is zero only at the origin. Consequently, equation (23) is a negative definite function. This completes the proof. $\square$

Robust backstepping control design

In this section, we consider a variable transformation as follows

$$\mathbf{Z} = \omega_e^r - \Phi(q_e) \quad (28)$$

In other words, according to equation (14), we assume that the actuated variable ω_e^r is outside the hypersurface Φ ; therefore, the transformed cascade model is written as follows

$$\dot{\mathbf{Z}} = \mathbf{u} + \mathbf{F} + \mathbf{d} - \frac{\partial \Phi(q_e)}{\partial q_e} \dot{q}_e \quad (29a)$$

$$\dot{q}_e = \frac{1}{2} \begin{bmatrix} q_{e0} & -q_{e3} \\ q_{e3} & q_{e0} \\ -q_{e2} & q_{e1} \end{bmatrix} (\mathbf{Z} + \Phi(q_e)) = \mathbf{A}(q_e)(\mathbf{Z} + \Phi) \quad (29b)$$

where $\mathbf{F}$ and $\mathbf{d}$ are defined in equation (11a) and (11b). Afterward, the control input is designed such that transformed cascade model of the underactuated spacecraft is asymptotically stable under the perturbations.

Theorem 2. Consider the underactuated spacecraft governed by equations (11a), (11b), (29a), and (29b) with FNDO given in equations (12) and (13), Remarks 1–7, Assumptions 1–4, and Corollary 1. $\mathbf{e}_{\chi_0}$ and $\mathbf{e}_{\chi_1}$ converge to zero in finite time by choice of $\Gamma_z \in R^{2 \times 2}$, κ_i , and L_i . So, under FNDO-BS controller proposed in equation (30), the state variables of the cascade model converge to the origin

$$\begin{aligned} \mathbf{u} = & -\mathbf{F} - \chi_1 + \frac{\partial \Phi(q_e)}{\partial q_e} \mathbf{A}(q_e)(\mathbf{Z} + \Phi) \\ & - (\mathbf{q}_e^T \mathbf{A} \mathbf{A}(q_e))^T - \Gamma_z \mathbf{Z} \end{aligned} \quad (30)$$

Proof. Firstly, according to Lemma 1, the Lyapunov function to analyze the stability of the whole system including the controller and FNDO is chosen as follows

$$V = V_a + \frac{1}{2} \mathbf{Z}^T \mathbf{Z} + V_{e_x} \quad (31)$$

According to equations (17), (29a), and (29b), we have

$$\begin{aligned} \dot{V} = & \mathbf{q}_e^T \mathbf{A} \mathbf{A}(q_e)(\mathbf{Z} + \Phi) \\ & + \mathbf{Z}^T \left\{ \mathbf{u} + \mathbf{F} + \mathbf{d} - \frac{\partial \Phi(q_e)}{\partial q_e} \mathbf{A}(q_e)(\mathbf{Z} + \Phi) \right\} + \dot{V}_{e_x} \end{aligned} \quad (32)$$

Equation (32) can be put in the following form

$$\begin{aligned} \dot{V} = & \mathbf{q}_e^T \mathbf{A} \mathbf{A}(q_e)(\Phi(q_e)) + \mathbf{q}_e^T \mathbf{A} \mathbf{A}(q_e) \mathbf{Z} \\ & + \mathbf{Z}^T \left\{ \mathbf{u} + \mathbf{F} + \mathbf{d} - \frac{\partial \Phi(q_e)}{\partial q_e} \mathbf{A}(q_e)(\mathbf{Z} + \Phi) \right\} + \dot{V}_{e_x} \end{aligned} \quad (33)$$

Given that $\mathbf{q}_e^T \mathbf{A} \mathbf{A}(q_e) \mathbf{Z}$ is a scalar term, equation (33) can be further written as follows

$$\begin{aligned} \dot{V} = & \mathbf{q}_e^T \mathbf{A} \mathbf{A}(q_e)(\Phi(q_e)) + \mathbf{Z}^T (\mathbf{q}_e^T \mathbf{A} \mathbf{A}(q_e))^T \\ & + \mathbf{Z}^T \left\{ \mathbf{u} + \mathbf{F} + \mathbf{d} - \frac{\partial \Phi(q_e)}{\partial q_e} \mathbf{A}(q_e)(\mathbf{Z} + \Phi) \right\} + \dot{V}_{e_x} \end{aligned} \quad (34)$$

Thus

$$\begin{aligned} \dot{V} = & \mathbf{q}_e^T \mathbf{A} \mathbf{A}(q_e)(\Phi(q_e)) \\ & + \mathbf{Z}^T \left\{ \mathbf{u} + \mathbf{F} + \mathbf{d} - \frac{\partial \Phi(q_e)}{\partial q_e} \mathbf{A}(q_e)(\mathbf{Z} + \Phi) \right. \\ & \left. + (\mathbf{q}_e^T \mathbf{A} \mathbf{A}(q_e))^T \right\} + \dot{V}_{e_x} \end{aligned} \quad (35)$$

Consequently, the control input for stabilizing the underactuated spacecraft is designed as follows

$$\begin{aligned} \mathbf{u} = & -\mathbf{F} - \chi_1 + \frac{\partial \Phi(q_e)}{\partial q_e} \mathbf{A}(q_e)(\mathbf{Z} + \Phi) \\ & - (\mathbf{q}_e^T \mathbf{A} \mathbf{A}(q_e))^T - \Gamma_z \mathbf{Z} \end{aligned} \quad (36)$$

where $\Gamma_z \in R^{2 \times 2}$ is the gain controller. Using equation (36) and substituting this into equation (35) yields

$$\begin{aligned} \dot{V} = & \mathbf{q}_e^T \mathbf{A} \mathbf{A}(q_e)(\Phi(q_e)) + \mathbf{Z}^T \{-\mathbf{e}_{\chi_1} - \Gamma_z \mathbf{Z}\} + \dot{V}_{e_x} \\ = & \mathbf{q}_e^T \mathbf{A} \mathbf{A}(q_e)(\Phi(q_e)) - \mathbf{Z}^T \mathbf{e}_{\chi_1} - \mathbf{Z}^T \Gamma_z \mathbf{Z} + \dot{V}_{e_x} \end{aligned} \quad (37)$$

Based on linear algebra, we know $-(1/2)(\|\mathbf{Z}\|^2 + \|\mathbf{e}_{\chi_1}\|^2) \leq \mathbf{Z}^T \mathbf{e}_{\chi_1} \leq (1/2)(\|\mathbf{Z}\|^2 + \|\mathbf{e}_{\chi_1}\|^2)$; then, equation (37) can be further written as follows

$$\begin{aligned} \dot{V} \leq & \mathbf{q}_e^T \mathbf{A} \mathbf{A}(q_e)(\Phi(q_e)) + \frac{1}{2} (\|\mathbf{Z}\|^2 + \|\mathbf{e}_{\chi_1}\|^2) \\ & - \mathbf{Z}^T \Gamma_z \mathbf{Z} + \dot{V}_{e_x} \end{aligned} \quad (38)$$

Taking into account Lemma 1 and $\lambda_{\min}(\Gamma_z) \|\mathbf{Z}\|^2 \leq \mathbf{Z}^T \Gamma_z \mathbf{Z} \leq \lambda_{\max}(\Gamma_z) \|\mathbf{Z}\|^2$, we further obtain

$$\begin{aligned} \dot{V} \leq & \mathbf{q}_e^T \mathbf{A} \mathbf{A}(q_e)(\Phi(q_e)) + \frac{1}{2} (\|\mathbf{Z}\|^2 + \|\mathbf{e}_{\chi_1}\|^2) \\ & - \mathbf{Z}^T \Gamma_z \mathbf{Z} - \beta_{\chi_0} \|\mathbf{e}_{\chi_0}\|^2 - \beta_{\chi_1} \|\mathbf{e}_{\chi_1}\|^2 \\ \leq & \mathbf{q}_e^T \mathbf{A} \mathbf{A}(q_e)(\Phi(q_e)) + \frac{1}{2} (\|\mathbf{Z}\|^2 + \|\mathbf{e}_{\chi_1}\|^2) \\ & - \lambda_{\min}(\Gamma_z) \|\mathbf{Z}\|^2 - \beta_{\chi_0} \|\mathbf{e}_{\chi_0}\|^2 - \beta_{\chi_1} \|\mathbf{e}_{\chi_1}\|^2 \end{aligned} \quad (39)$$

Finally, we have

$$\dot{V} \leq \mathbf{q}_e^T \mathbf{A} \mathbf{A}(\mathbf{q}_e)(\Phi(\mathbf{q}_e)) - \beta_{x_0} \|\mathbf{e}_{x_0}\|^2 - \|\mathbf{Z}\|^2 \left(\lambda_{\min}(\Gamma_z) - \frac{1}{2} \right) - \|\mathbf{e}_{x_1}\|^2 \left(\beta_{x_1} - \frac{1}{2} \right) \quad (40)$$

Moreover, with regard to the result of Theorem 1, $\mathbf{q}_e^T \mathbf{A} \mathbf{A}(\mathbf{q}_e)(\Phi(\mathbf{q}_e))$ is a negative definite function. Also, by choosing $\beta_{x_1} \lambda_{\min}(\Gamma_z) > (1/2)$, $\dot{V}$ is a negative definite function, so Theorems 1 and 2 provide stability of FNDO-BS controller asymptotically under time-varying perturbations. This completes the proof.

Remark 9. Using equation (20), $(\partial\Phi(\mathbf{q}_e)/\partial\mathbf{q}_e) \in R^{2 \times 3}$ is obtained as follows

$$\frac{\partial\Phi(\mathbf{q}_e)}{\partial\mathbf{q}_e} = \begin{bmatrix} -2\lambda_1 q_{e0} & -2a q_{e3} & -2a q_{e2} \\ -2b q_{e3} & -2\lambda_2 q_{e0} & -2b q_{e1} \end{bmatrix} \quad (41)$$

Remark 10. Given equation (31), V is radially unbounded function; then, the closed-loop system has global asymptotic stability because of equation (41).

Modification for saturation consideration

The saturation occurrence is one of the major problems of control systems that have to be considered in the design process. The saturation can destabilize the closed-loop system; therefore, controller design with consideration of the actuator saturation is a mandatory step. Although the control design under the saturation has been reported,^{12,30,39,40} the attitude tracking control of the underactuated spacecraft with the input saturation is not enough studied. Consider $\mathbf{sat}(\mathbf{u}) = [\text{sat}(u_x) \text{ sat}(u_y)]^T$ is the actual control input generated by reaction wheels.

Remark 11. As known, $\text{sat}(u_j), j = x, y$, denotes the nonlinear property of the actuator and can be expressed as

$$\text{sat}(u_j) = \begin{cases} u_j, & |u_j| \leq u_s \\ \text{sign}(u_j) u_s, & |u_j| > u_s \end{cases} \quad (42)$$

where u_s is a saturation level. From equation (10a) and (10b), we rewrite the cascade model with the actuator saturation as follows

$$\dot{\omega}_e^r = \mathbf{sat}(\mathbf{u}) + \mathbf{F} + \mathbf{d} \quad (43a)$$

$$\dot{\mathbf{q}}_e = \frac{1}{2} \begin{bmatrix} q_{e0} & -q_{e3} \\ q_{e3} & q_{e0} \\ -q_{e2} & q_{e1} \end{bmatrix} \omega_e^r \quad (43b)$$

where $\mathbf{F}$ is obtained from equation (11a). Given Remark 3 and equation (11b), $\mathbf{d}$ can be put in the following form

$$\mathbf{d} = -\rho_w \mathbf{sat}(\mathbf{u}) - \overline{\Delta \mathbf{J}^r} \{ \mathbf{I}_2 - \rho_w \} \mathbf{J}_0^r \mathbf{sat}(\mathbf{u}) + \{ \mathbf{J}_0^r + \Delta \mathbf{J}^r \}^{-1} \mathbf{T}_d^r \quad (44)$$

In order to reduce the saturation effect, FNDO-BS is modified by Theorem 3 to enhance the robustness against input saturation.

Assumption 5. Consider $\mathbf{sat}_\delta(\mathbf{Y}) = [\text{sat}_\delta(y_1) \dots \text{sat}_\delta(y_n)]^T$ for every $\mathbf{Y} \in R^n$, where $|\text{sat}_\delta(y_j)| \leq 1$ and $\text{sat}_\delta(y_j)$ is defined as

$$\text{sat}_\delta(y_j) = \begin{cases} y_j, & |y_j| \leq 1 \\ \text{sign}(y_j), & |y_j| > 1 \end{cases} \quad (45)$$

Theorem 3. For the cascade model governed by equation (43a) and (43b) under the actuator saturation and the variable transformation proposed in equation (28), with FNDO given in equation (12) to reject total perturbations, there exist an auxiliary vector $\mu \in R^2$ with $\alpha_i > 0$, κ_i , and L_i such that $\mathbf{e}_{x_0}$ and $\mathbf{e}_{x_1}$ converge to zero in finite time, and cascade model variables are asymptotically stable by using the modification of the control input $\mathbf{u}$ as follows

$$\mathbf{u} = -\alpha_1 \mathbf{sat}_\delta(\mu) - \alpha_2 \mathbf{sat}_\delta(\mathbf{Z}) \quad (46)$$

where α_1 and α_2 are constant gain, and μ is chosen as

$$\dot{\mu} = \frac{\mu \mathbf{Z}^T}{\|\mu\|^2} \left\{ \alpha_1 \mathbf{sat}_\delta(\mu) - \mathbf{F} - \chi_1 + \frac{\partial\Phi(\mathbf{q}_e)}{\partial\mathbf{q}_e} \mathbf{A}(\mathbf{q}_e)(\mathbf{Z} + \Phi) - (\mathbf{q}_e^T \mathbf{A} \mathbf{A}(\mathbf{q}_e))^T - \Gamma_z \mathbf{Z} \right\} - \Gamma_\mu \mu \quad (47)$$

where $\Gamma_\mu \in R^{3 \times 3}$ and $\Gamma \in R^{3 \times 3}$ are diagonal positive definite matrices. Obviously, the equilibrium point of equation (47) is zero.

Proof. According to Lemma 1, the Lyapunov function to analyze the stability of the whole system including controller and MFNDO is chosen as follows

$$V_1 = V_a + \frac{1}{2} \mathbf{Z}^T \mathbf{Z} + \frac{1}{2} \mu^T \mu + V_{e_x} \quad (48)$$

Consider equations (28), (35), (43a), and (46); thus, $\dot{V}_1$ can be put in the following form

$$\begin{aligned} \dot{V}_1 &= \dot{V}_a + \mathbf{Z}^T \dot{\mathbf{Z}} + \mu^T \dot{\mu} + \dot{V}_{e_x} \\ &= \mathbf{q}_e^T \mathbf{A} \mathbf{A}(\mathbf{q}_e)(\Phi(\mathbf{q}_e)) + \mathbf{Z}^T \left\{ -\alpha_1 \mathbf{sat}_\delta(\mu) - \alpha_2 \mathbf{sat}_\delta(\mathbf{Z}) \right. \\ &\quad \left. + \mathbf{F} + \mathbf{d} - \frac{\partial\Phi(\mathbf{q}_e)}{\partial\mathbf{q}_e} \mathbf{A}(\mathbf{q}_e)(\mathbf{Z} + \Phi) + (\mathbf{q}_e^T \mathbf{A} \mathbf{A}(\mathbf{q}_e))^T \right\} \\ &\quad + \mu^T \dot{\mu} + \dot{V}_{e_x} \end{aligned} \quad (49)$$

Using equations (47) and (49) and Lemma 1 yields

$$\begin{aligned}
\dot{V}_1 &= \mathbf{q}_e^T \mathbf{A} \mathbf{A}(\mathbf{q}_e)(\Phi(\mathbf{q}_e)) + \mathbf{Z}^T \left\{ -\alpha_1 \text{sat}_\delta(\boldsymbol{\mu}) - \alpha_2 \text{sat}_\delta(\mathbf{Z}) + \mathbf{F} + \mathbf{d} - \frac{\partial \Phi(\mathbf{q}_e)}{\partial \mathbf{q}_e} \mathbf{A}(\mathbf{q}_e)(\mathbf{Z} + \Phi) + (\mathbf{q}_e^T \mathbf{A} \mathbf{A}(\mathbf{q}_e))^T \right\} \\
&\quad + \boldsymbol{\mu}^T \left(\frac{\boldsymbol{\mu} \mathbf{Z}^T}{\|\boldsymbol{\mu}\|^2} \left\{ \alpha_1 \text{sat}_\delta(\boldsymbol{\mu}) - \mathbf{F} - \chi_1 + \frac{\partial \Phi(\mathbf{q}_e)}{\partial \mathbf{q}_e} \mathbf{A}(\mathbf{q}_e)(\mathbf{Z} + \Phi) - (\mathbf{q}_e^T \mathbf{A} \mathbf{A}(\mathbf{q}_e))^T - \Gamma_z \mathbf{Z} \right\} - \Gamma_\mu \boldsymbol{\mu} \right) + \dot{V}_{e_x} \\
&\leq \mathbf{q}_e^T \mathbf{A} \mathbf{A}(\mathbf{q}_e)(\Phi(\mathbf{q}_e)) + \mathbf{Z}^T \left\{ -\alpha_1 \text{sat}_\delta(\boldsymbol{\mu}) - \alpha_2 \text{sat}_\delta(\mathbf{Z}) + \mathbf{F} + \mathbf{d} - \frac{\partial \Phi(\mathbf{q}_e)}{\partial \mathbf{q}_e} \mathbf{A}(\mathbf{q}_e)(\mathbf{Z} + \Phi) + (\mathbf{q}_e^T \mathbf{A} \mathbf{A}(\mathbf{q}_e))^T \right\} \\
&\quad + \left(\mathbf{Z}^T \left\{ \alpha_1 \text{sat}_\delta(\boldsymbol{\mu}) - \mathbf{F} - \chi_1 + \frac{\partial \Phi(\mathbf{q}_e)}{\partial \mathbf{q}_e} \mathbf{A}(\mathbf{q}_e)(\mathbf{Z} + \Phi) - (\mathbf{q}_e^T \mathbf{A} \mathbf{A}(\mathbf{q}_e))^T - \Gamma_z \mathbf{Z} \right\} - \boldsymbol{\mu}^T \Gamma_\mu \boldsymbol{\mu} \right) + \dot{V}_{e_x} \\
&\leq \mathbf{q}_e^T \mathbf{A} \mathbf{A}(\mathbf{q}_e)(\Phi(\mathbf{q}_e)) + \mathbf{Z}^T \{ -\alpha_2 \text{sat}_\delta(\mathbf{Z}) - \mathbf{e}_{\chi_1} - \Gamma_z \mathbf{Z} \} - \boldsymbol{\mu}^T \Gamma_\mu \boldsymbol{\mu} - \beta_{\chi_0} \|\mathbf{e}_{\chi_0}\|^2 - \beta_{\chi_1} \|\mathbf{e}_{\chi_1}\|^2
\end{aligned} \tag{50}$$

Finally, due to $\lambda_{\min}(\Gamma_z) \|\mathbf{Z}\|^2 \leq \mathbf{Z}^T \Gamma_z \mathbf{Z} \leq \lambda_{\max}(\Gamma_z) \|\mathbf{Z}\|^2$, we easily have

$$\begin{aligned}
\dot{V}_1 &\leq \mathbf{q}_e^T \mathbf{A} \mathbf{A}(\mathbf{q}_e)(\Phi(\mathbf{q}_e)) - \alpha_2 \mathbf{Z}^T \text{sat}_\delta(\mathbf{Z}) - \boldsymbol{\mu}^T \Gamma_\mu \boldsymbol{\mu} \\
&\quad - \beta_{\chi_0} \|\mathbf{e}_{\chi_0}\|^2 - \|\mathbf{Z}\|^2 \left(\lambda_{\min}(\Gamma_z) - \frac{1}{2} \right) \\
&\quad - \|\mathbf{e}_{\chi_1}\|^2 \left(\beta_{\chi_1} - \frac{1}{2} \right)
\end{aligned} \tag{51}$$

As mentioned before, $\mathbf{q}_e^T \mathbf{A} \mathbf{A}(\mathbf{q}_e)(\Phi(\mathbf{q}_e))$ is a negative definite function. Also, by choosing $\beta_{\chi_1}, \lambda_{\min}(\Gamma_z) > (1/2)$; it is clear that V_1 is a negative definite function. Therefore, all state variables of the closed-loop cascade model and also the auxiliary vector $\boldsymbol{\mu}$ are converged to zero asymptotically. This completes the proof. $\square$

Remark 12. According to equation (48), V_1 is radially unbounded function; then, the closed-loop system has global asymptotic stability because of equation (51).

Remark 13. Due to equation (46), the actuator saturation can be attenuated by proper choice of parameters v_1 and v_2 .

Selecting small amounts will reduce the quality of the controller, while larger values of parameters cause the actuator to be saturated; thus, there exist a trade-off between the saturation reduction and the control performance.

Verification by Adams-MATLAB co-simulation

At first, we constructed the spacecraft equipped with two reaction wheels in Adams/View software. Figure 1 shows the Adams model.

Then, in order to export Adams model (plant), inputs and outputs of the plant were identified by Adams/plugin-controls. Next, the closed-loop system was completed by importing the Adams model into MATLAB/Simulink environment. Figure 2 shows the block diagram of the co-simulation logic. The calculated inputs in MATLAB/Simulink which must be generated by reaction wheels were simultaneously applied to the Adams model.

According to the input signals, the Adams model rotated, and then, the output signals, the angular velocities, and the quaternions were produced based on the motion of

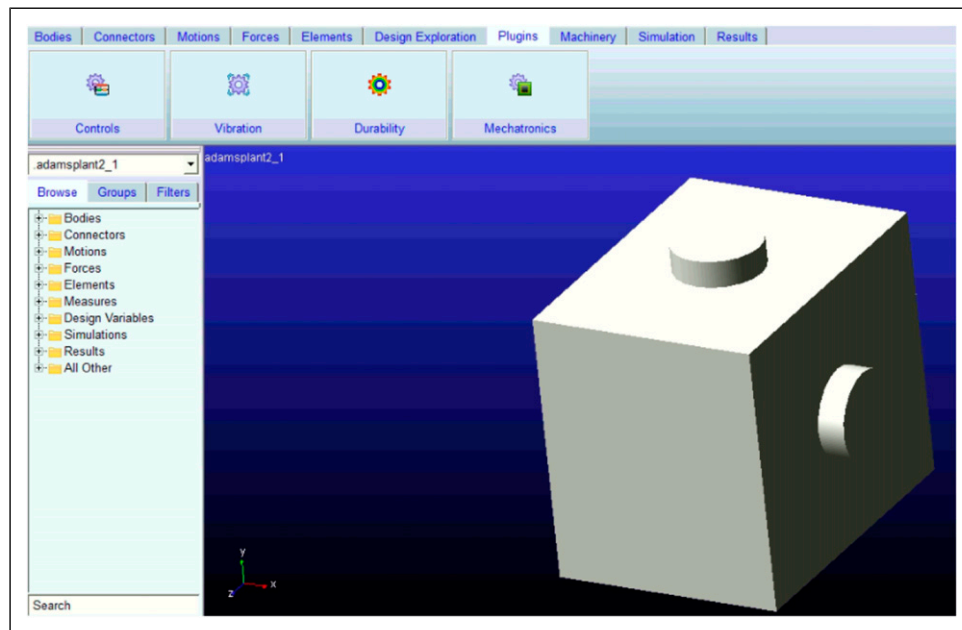


Figure 1. Adams model of spacecraft equipped with two reaction wheels.

the Adams model. Output signals were imported to the controller in MATLAB/Simulink. Therefore, the next iteration could be repeated similarly. For verification, we considered two simulation scenarios:

MATLAB scenario: Both plant and controller are in MATLAB/Simulink environment.

Adams scenario: According to the above description, Adams-MATLAB co-simulation is considered.

Therefore, due to scenarios, we have plotted quaternions and angular velocities in Figures 3 and 4. As we can see, the maximum error between Adams and MATLAB scenarios for quaternions is less than 19%, while the maximum error for angular velocities is less than 13.5%.

Results and discussion

In this section, in order to evaluate the stability and the robustness, the designed controllers of the under-actuated spacecraft were simulated under the large-angle attitude tracking maneuver. For all simulations,

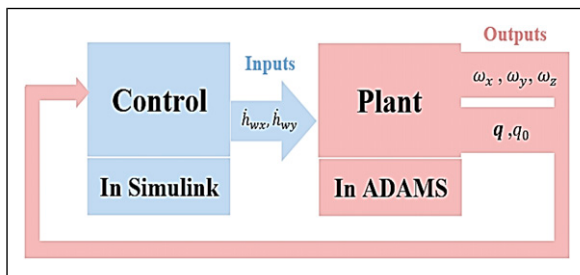


Figure 2. Co-simulation diagram.

values of J_0 , A , Γ_z , Γ_μ , L , κ_0 , κ_1 , α_1 , and α_2 are assumed to be

$$\begin{bmatrix} 2.5 & 0 & 0 \\ 0 & 3.5 & 0 \\ 0 & 0 & 4.5 \end{bmatrix} (\text{Kg m}^2), \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 6 \end{bmatrix}, 1.5 \begin{bmatrix} 1 & 0 \\ 0 & 0.8 \end{bmatrix},$$

$$0.5 \begin{bmatrix} 0.1 & 0 \\ 0 & 0.1 \end{bmatrix}, 10 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, 0.1, 0.15, 0.2, \text{ and } 0.3,$$

respectively. Given perturbations, values of ΔJ and ρ_w are equal to $\text{diag}(2 \sin(0.1t))$, $2.8 \sin(0.2t)$, $3.6 \sin(0.3t)$, and $0.35I_2$, respectively. Furthermore, it is assumed the external disturbances to be summation of constant and harmonic functions as

$$T_d^r = 7.5 \times 10^{-3} [0.1 + 9 \sin(0.5t) \quad 0.1 + 7.5 \sin(0.8t)]^T (\text{Nm})$$

For the purpose of comparison, backstepping (BS), FNDO-BS, and MFNDO-BS controllers are applied to the problem. Note that the BS controller has a structure similar to equation (30) whose disturbance observer introduced in Lemma 1 was turned off.

Attitude tracking control in the presence of time-varying perturbations

In this step, we first compare simulation results of MFNDO-BS (solid lines), FNDO-BS (dashed lines), and BS (dotted line) for the attitude tracking control of the underactuated spacecraft with perturbations under the large-angle maneuver. Figure 5 shows that the large-angle attitude maneuver is more than 90° . This condition is applied throughout the article. Notice that the saturation is not considered in this case.

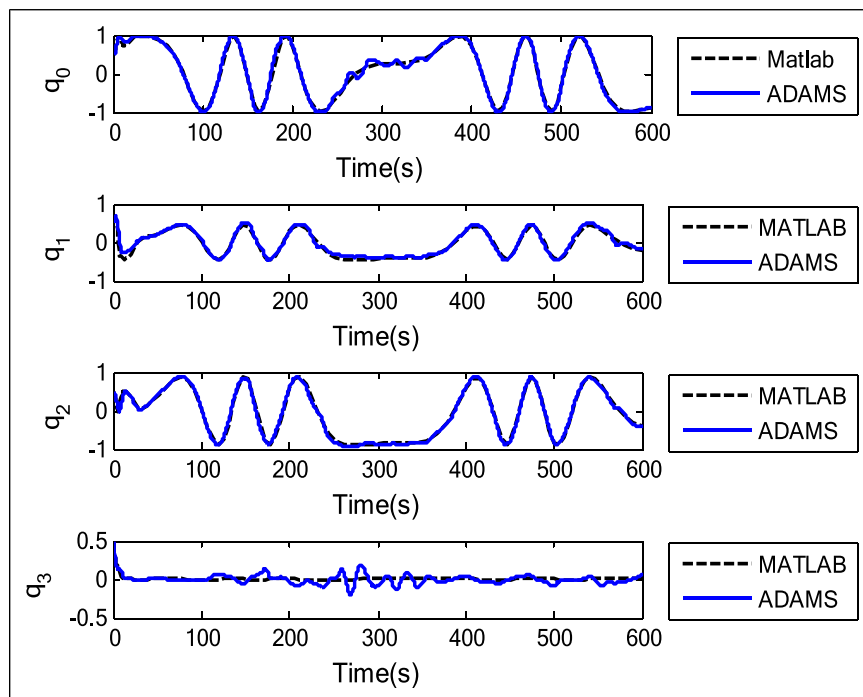


Figure 3. Comparing between Adams and MATLAB for quaternions.

As shown in Figures 6 and 7, although figures have transitional fluctuation, the errors of the quaternions and the angular velocities converge to zero within 60 s.

However, both MFNDO-BS and FNDO-BS controllers under time-varying perturbation have the same behavior; obviously, the BS controller cannot converge to origin asymptotically in this case. According to

Figure 8, in the first 20 s, both FNDO-BS and BS controllers have a control input of about 10 Nm, while the control input of MFNDO-BS is 0.5 Nm over the same time range.

Moreover, Figure 9 shows comparing estimation errors of total perturbations between MFNDO-BS and FNDO-BS controllers.

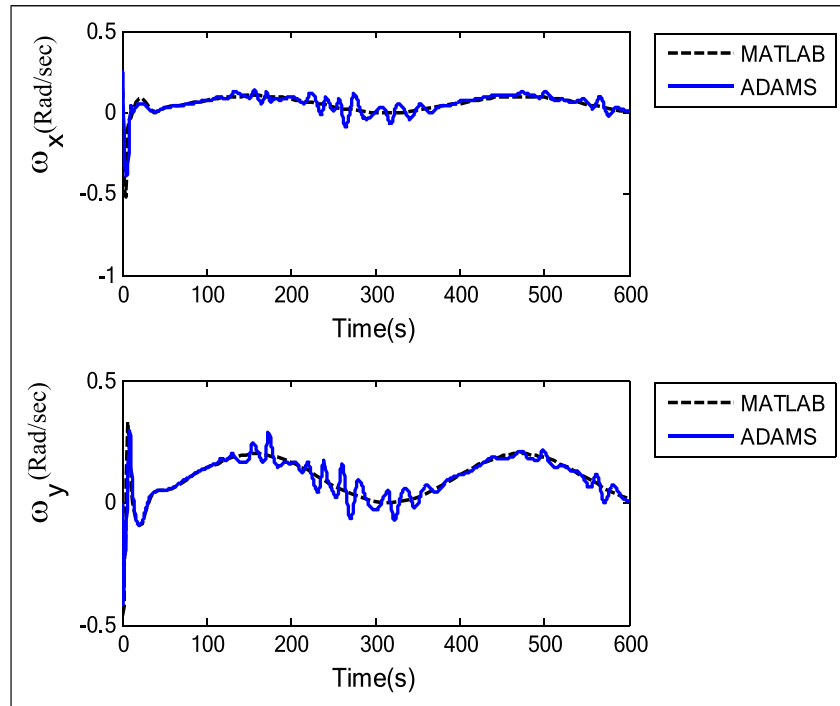


Figure 4. Comparing between Adams and MATLAB for angular velocities.

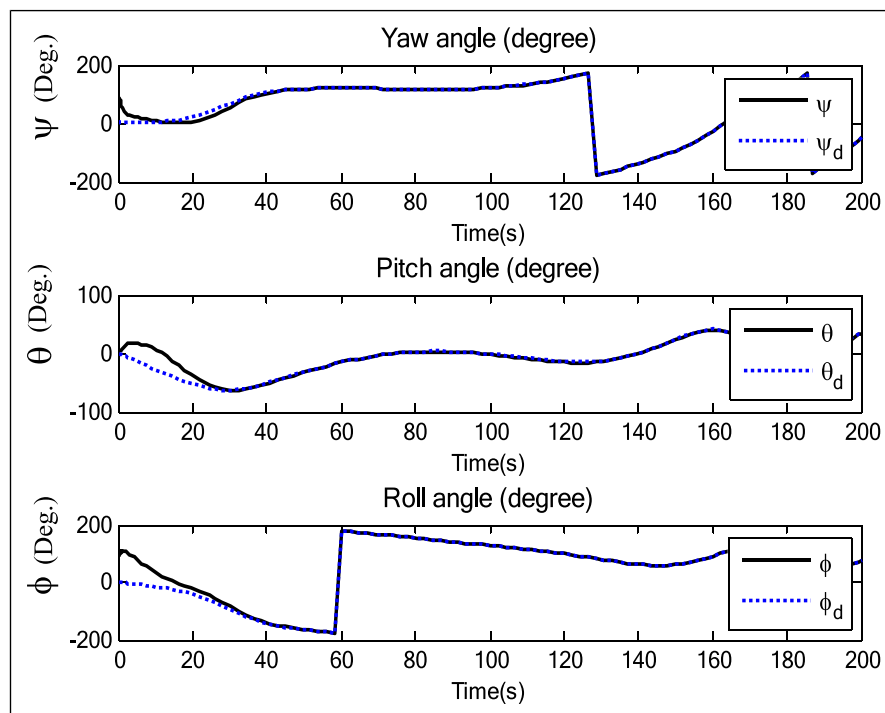


Figure 5. Comparing between the Euler angle parameters and their desired path.

Attitude tracking control in the presence of time-varying perturbations and saturation

In this subsection, the performance of MFNDO-BS (solid lines), FNDO-BS (dashed lines), and BS (dotted line) are compared in the same maneuver with saturation and perturbations consideration. Values of parameters and perturbations are mentioned before. We consider two

saturation conditions to evaluate MFNDO-BS, FNDO-BS, and BS.

Condition I. Soft saturation

In this case, the restriction on the actuator torque magnitude with -1 Nm and 1 Nm are presumed. As from Figure 10, we can see FNDO-BS and BS controllers cannot even be

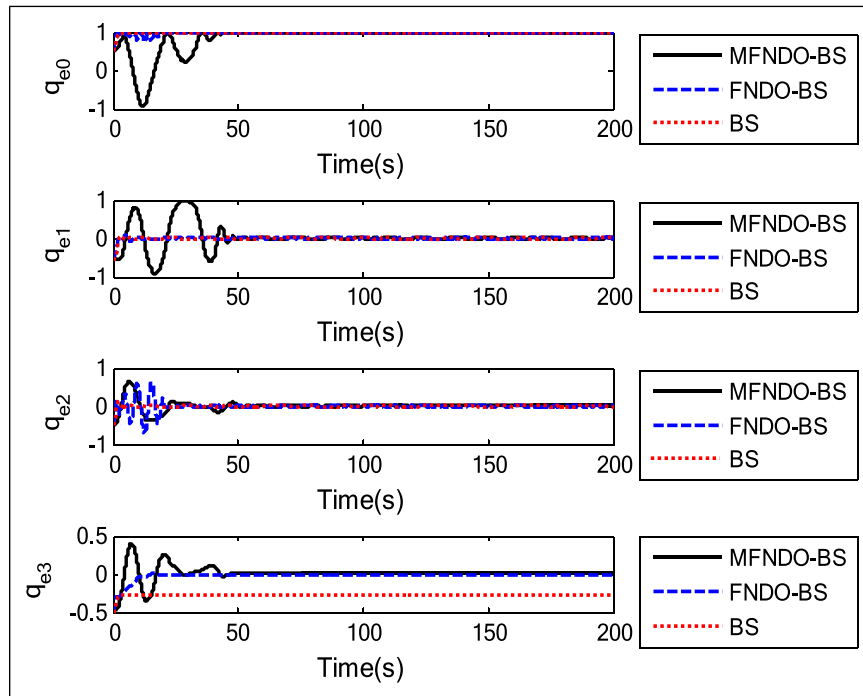


Figure 6. Comparing the quaternion errors for the attitude tracking control with the perturbations.

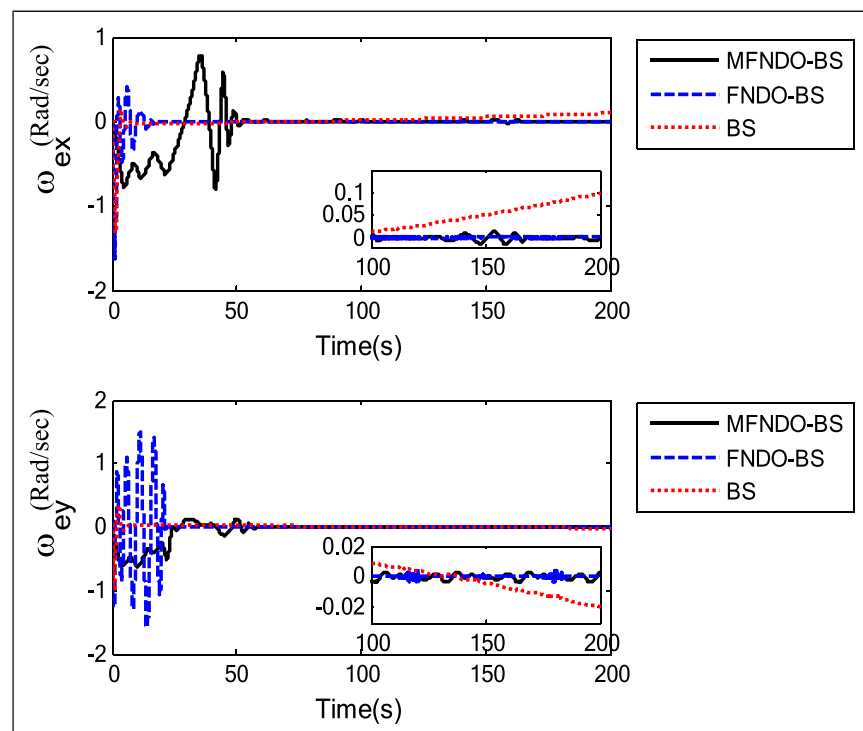


Figure 7. Comparing the angular velocity errors for the attitude tracking control with the perturbations.

stable under the soft saturation. In this case, the MFNDO-BS controller is asymptotically stable so that quaternions and angular velocities converge to the desired path within 80 s. In fact, by modifying FNDO-BS scheme, MFNDO-BS is sufficiently robust against the soft saturation.

Condition II. Hard saturation

In the critical condition, we suppose that a weaker actuator than before with a saturation level equals to 0.035 Nm, 3.5% of the soft saturation level, is available. As from

Figure 11, it can be easily seen that MFNDO-BS is still asymptotic stable under the hard saturation.

Due to Figure 11, quaternion and angular velocity errors are converged to the origin within 150 s, although, as we can see, the actuator works near the saturation level. Consequently, as we can see, the MFNDO-BS controller has sufficient robustness against the hard saturation.

Also, Figure 12 shows the simulation of the perturbation $\mathbf{d}$ (see equation (44)), the disturbance estimation error $\mathbf{e}_{\chi 1}$ (see equation (13)), and the auxiliary vector $\boldsymbol{\mu}$

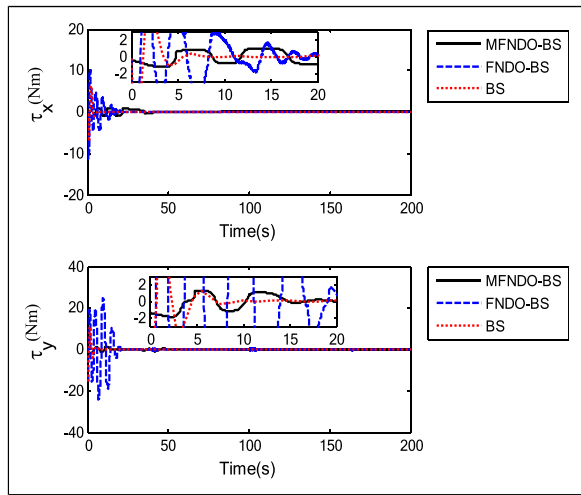


Figure 8. Comparing the control inputs for the attitude tracking control with the perturbations.

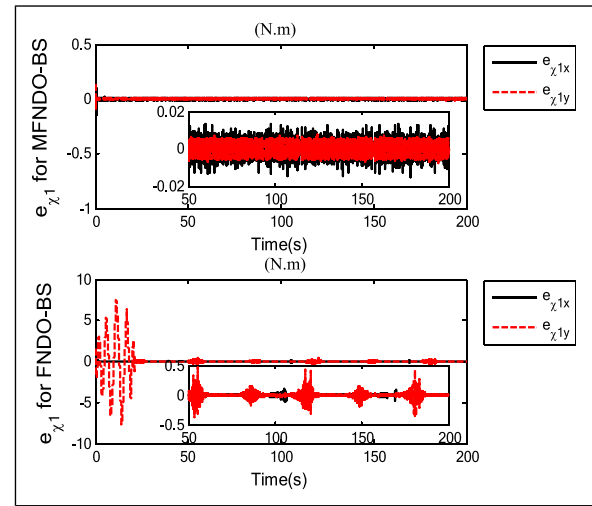


Figure 9. Comparing the errors of the disturbance estimation.

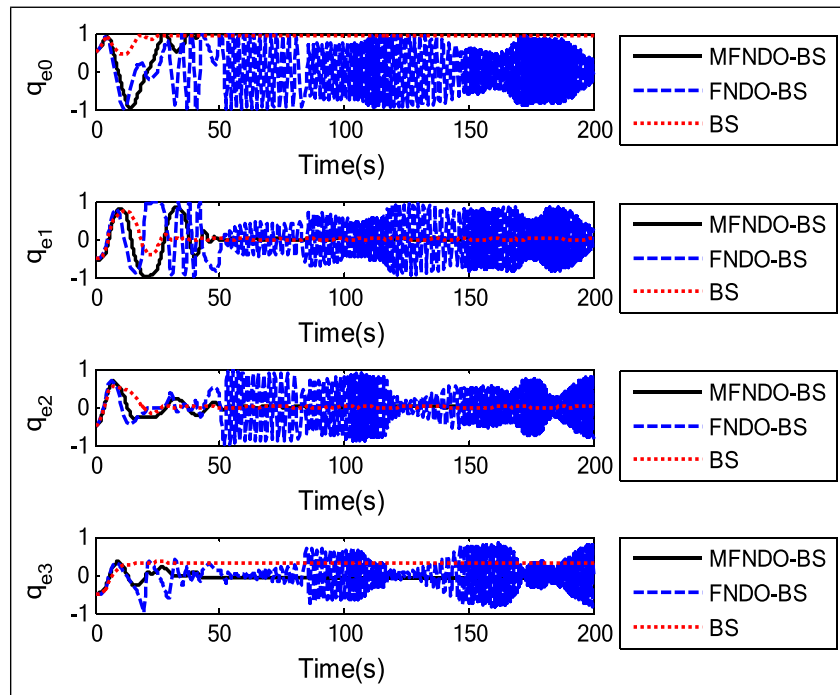


Figure 10. Comparing the quaternion errors in the presence of the perturbations and soft saturation.

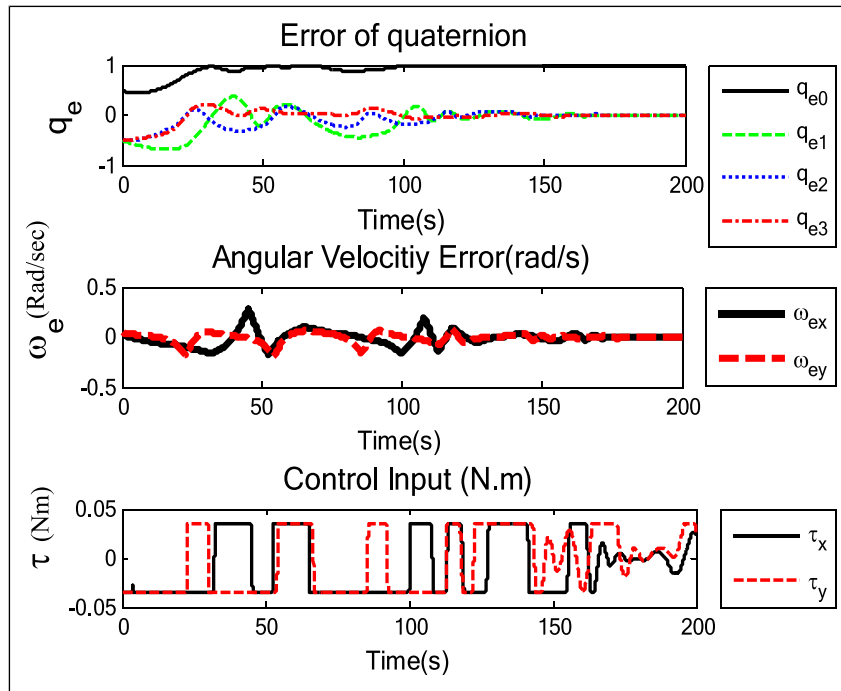


Figure 11. q_e , ω_e , and control input of modified FNDO-BS under perturbations and hard saturation.

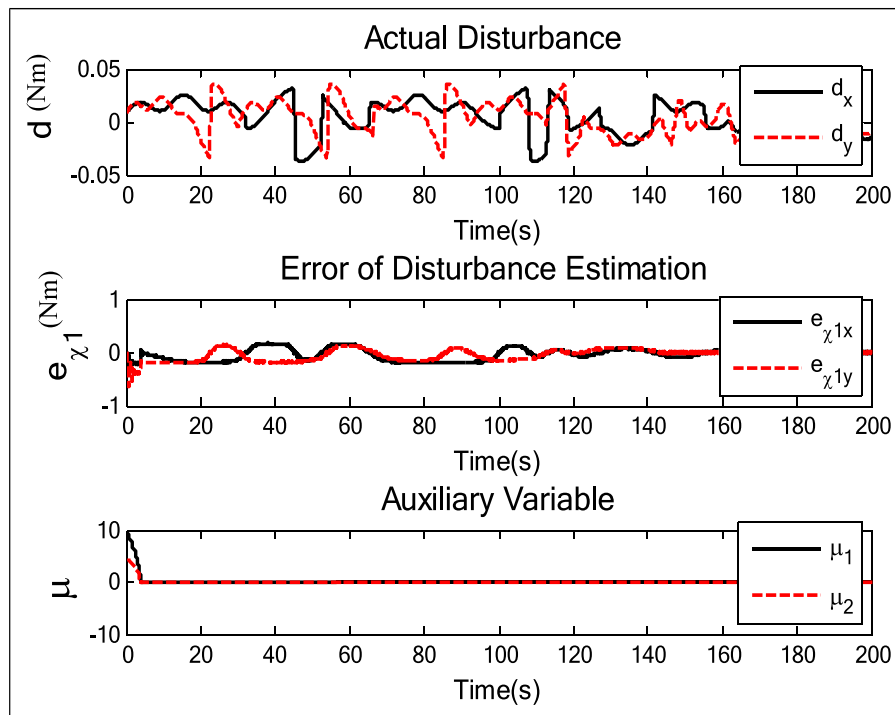


Figure 12. d , $e_{\chi 1}$, and μ of modified FNDO-BS under perturbations and hard saturation.

(see equation (47)) in the presence of hard saturation for MFNDO-BS controller; it is clear $e_{\chi 1}$ and μ are stable.

Conclusions

This study aims to design a robust and stable controller for underactuated spacecraft in a large attitude maneuver in presence of the saturation and time-varying perturbations.

The proposed controller utilizes a cascade model to achieve the perfect tracking which does not need limiting initial conditions of the quaternions because it is not singular. In order to enhance the robustness of the controller against external disturbances, inertia uncertainties, and internal disturbances of the actuators, all of which are time-variant; a finite-time nonlinear disturbance observer-based backstepping (FNDO-BS) was designed, and

Lyapunov criteria showed that the FNDO-BS scheme was asymptotically stable. Furthermore, MFNDO-BS was presented, which was asymptotically stable and had sufficient robustness in the presence of the saturation and upper-bounded time-varying perturbations. From the point view of practice, the MFNDO-BS controller had the asymptotic stability with adequate performance under the combination of saturation and uncertainty. Consequently, according to Figure 6–12, advantages of MFNDO-BS as the robust backstepping controller are achieving the asymptotic stability for the underactuated spacecraft with two reaction wheels and achieving the robustness in presence of external disturbances, inertia uncertainties, and internal disturbances of actuators and saturation. As mentioned previously, MFNDO-BS was asymptotically stable under the soft saturation. Furthermore, MFNDO-BS was asymptotic stable under the hard saturation whose level is equal to 0.035 Nm, 3.5% of the soft saturation level. The closed-loop system was verified by Adams-MATLAB/Simulink co-simulation. The maximum verification error for quaternions was less than 19%, while the maximum verification error for angular velocities was less than 13.5%.

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